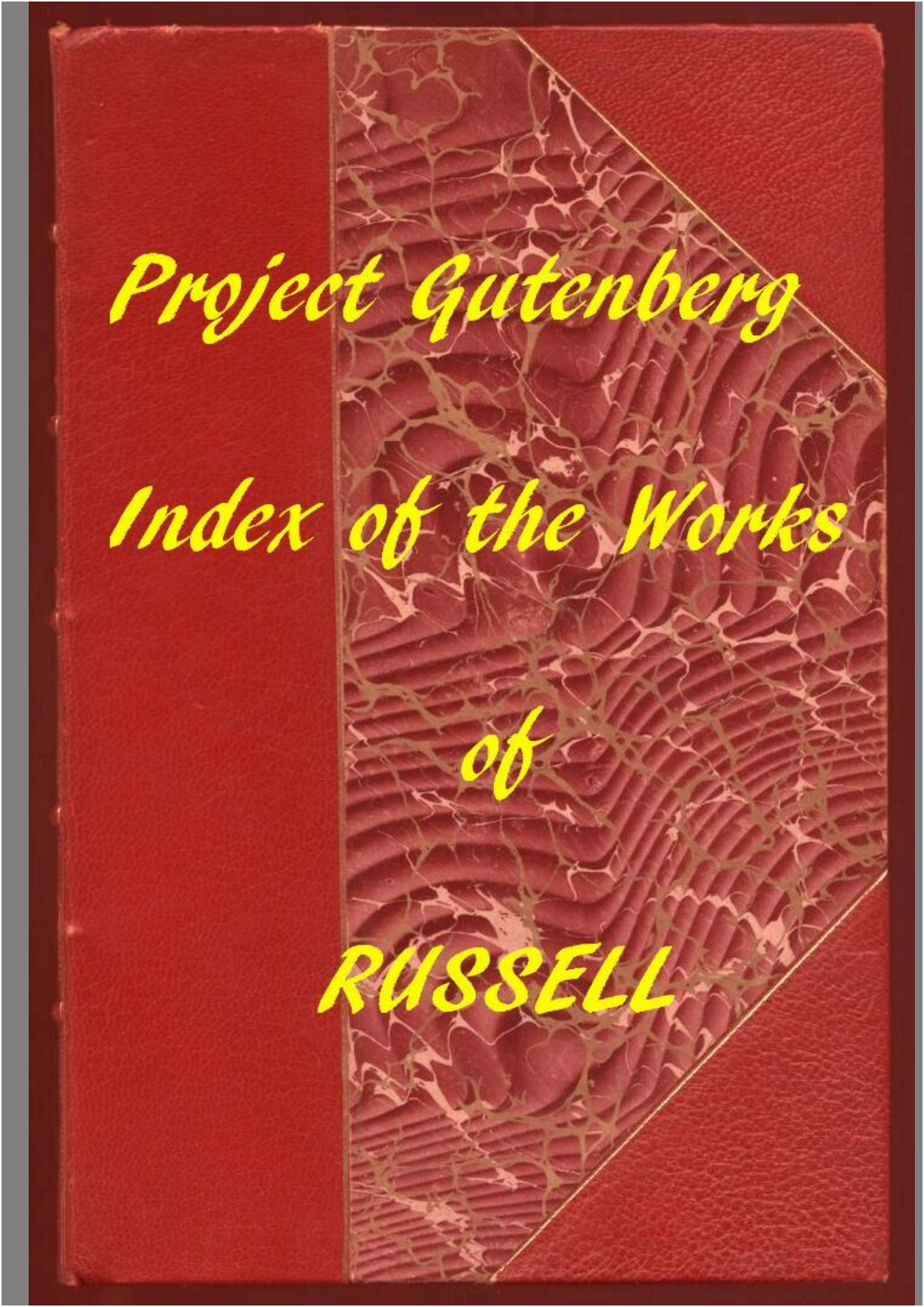


Project Gutenberg

Index of the Works

of

RUSSELL

The image shows the front cover of a book. The cover is primarily a deep red color with a fine, pebbled texture. A vertical strip of marbled paper runs down the center, and a diagonal section of marbled paper extends from the top right corner towards the center. The marbling features a complex, organic pattern of swirling, cell-like shapes in shades of brown, tan, and cream. The text is printed in a yellow, cursive-style font.

Project Gutenberg

Index of the Works

of

RUSSELL

The Project Gutenberg eBook of Index of the Project Gutenberg Works of Bertrand Russell

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Index of the Project Gutenberg Works of Bertrand Russell

Author: Bertrand Russell

Editor: David Widger

Release date: April 29, 2019 [eBook #59391]

Most recently updated: April 9, 2023

Language: English

Other information and formats: www.gutenberg.org/ebooks/59391

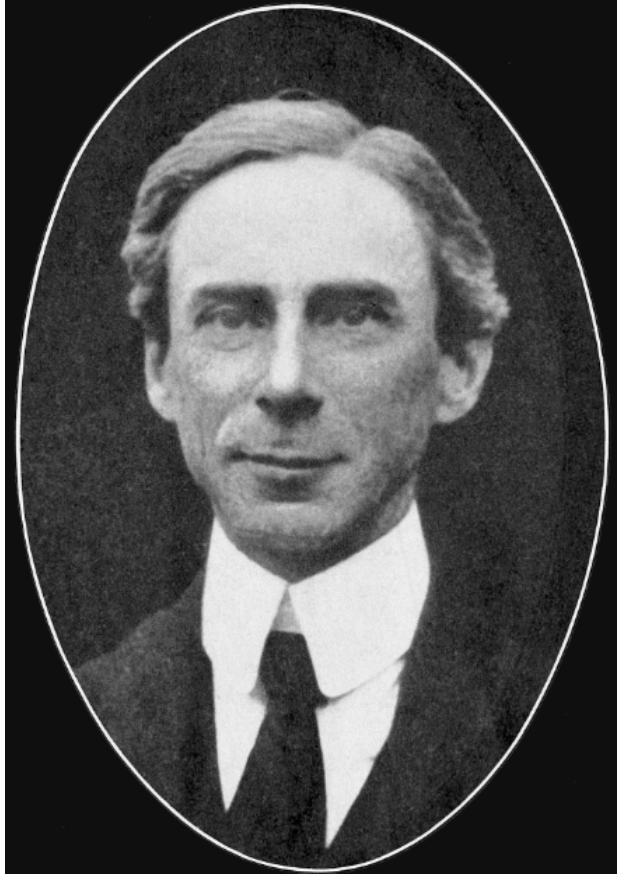
Credits: Produced by David Widger

*** START OF THE PROJECT GUTENBERG EBOOK INDEX OF THE
PROJECT GUTENBERG WORKS OF BERTRAND RUSSELL ***

**INDEX OF THE PROJECT
GUTENBERG**

**WORKS OF
BERTRAND RUSSELL**

Compiled by David Widger



CONTENTS

Click on the **##** before many of the titles to view a linked table of contents for that volume.

Click on the title itself to open the original online file.

PROPOSED ROADS TO FREEDOM

THE ANALYSIS OF MIND

POLITICAL IDEALS

THE PROBLEMS OF PHILOSOPHY

THE PROBLEM OF CHINA

BOLSHEVISM

MYSTICISM AND LOGIC

OUR KNOWLEDGE OF EXTERNAL WORLD

FREE THOUGHT AND OFFICIAL PROPAGANDA

ON FOUNDATIONS OF GEOMETRY

WHY MEN FIGHT

TABLES OF CONTENTS OF VOLUMES

PROPOSED ROADS TO FREEDOM

By Bertrand Russell

CONTENTS

[INTRODUCTION](#)

[PART I](#) HISTORICAL

[CHAPTER I](#) MARX AND SOCIALIST DOCTRINE

[CHAPTER II](#) BAKUNIN AND ANARCHISM

[CHAPTER III](#) THE SYNDICALIST REVOLT

[PART II](#) PROBLEMS OF THE FUTURE

[CHAPTER IV](#) WORK AND PAY

[CHAPTER V](#) GOVERNMENT AND LAW

[CHAPTER VI](#) INTERNATIONAL RELATIONS

[CHAPTER VII](#) SCIENCE AND ART UNDER SOCIALISM

[CHAPTER VIII](#) THE WORLD AS IT COULD BE MADE

THE ANALYSIS OF MIND

By Bertrand Russell

1921

CONTENTS

MUIRHEAD LIBRARY OF PHILOSOPHY

PREFACE

THE ANALYSIS OF MIND

<u>LECTURE I.</u>	RECENT CRITICISMS OF "CONSCIOUSNESS"
<u>LECTURE II.</u>	INSTINCT AND HABIT
<u>LECTURE III.</u>	DESIRE AND FEELING
<u>LECTURE IV.</u>	INFLUENCE OF PAST HISTORY ON PRESENT OCCURRENCES IN LIVING
<u>LECTURE V.</u>	PSYCHOLOGICAL AND PHYSICAL CAUSAL LAWS
<u>LECTURE VI.</u>	INTROSPECTION
<u>LECTURE VII.</u>	THE DEFINITION OF PERCEPTION
<u>LECTURE VIII.</u>	SENSATIONS AND IMAGES
<u>LECTURE IX.</u>	MEMORY
<u>LECTURE X.</u>	WORDS AND MEANING
<u>LECTURE XI.</u>	GENERAL IDEAS AND THOUGHT
<u>LECTURE XII.</u>	BELIEF
<u>LECTURE XIII.</u>	TRUTH AND FALSEHOOD
<u>LECTURE XIV.</u>	EMOTIONS AND WILL

[LECTURE XV.](#) CHARACTERISTICS OF MENTAL PHENOMENA

POLITICAL IDEALS

By Bertrand Russell

CONTENTS

- I: [Political Ideals](#)
 - II: [Capitalism and the Wage System](#)
 - III: [Pitfalls in Socialism](#)
 - IV: [Individual Liberty and Public Control](#)
 - V: [National Independence and Internationalism](#)
-

THE PROBLEMS OF PHILOSOPHY

By Bertrand Russell

CONTENTS

[PREFACE](#)

[CHAPTER I.](#)

APPEARANCE AND REALITY

[CHAPTER II.](#)

THE EXISTENCE OF MATTER

[CHAPTER III.](#)

THE NATURE OF MATTER

[CHAPTER IV.](#)

IDEALISM

[CHAPTER V.](#)

KNOWLEDGE BY ACQUAINTANCE AND
KNOWLEDGE BY DESCRIPTION

[CHAPTER VI.](#)

ON INDUCTION

[CHAPTER VII.](#)

ON OUR KNOWLEDGE OF GENERAL
PRINCIPLES

[CHAPTER VIII.](#)

HOW A *PRIORI* KNOWLEDGE IS
POSSIBLE

[CHAPTER IX.](#)

THE WORLD OF UNIVERSALS

[CHAPTER X.](#)

ON OUR KNOWLEDGE OF UNIVERSALS

[CHAPTER XI.](#)

ON INTUITIVE KNOWLEDGE

[CHAPTER XII.](#)

TRUTH AND FALSEHOOD

[CHAPTER XIII.](#)

KNOWLEDGE, ERROR, AND PROBABLE
OPINION

[CHAPTER XIV.](#)

THE LIMITS OF PHILOSOPHICAL
KNOWLEDGE

[CHAPTER XV.](#)

THE VALUE OF PHILOSOPHY

[BIBLIOGRAPHICAL
NOTE](#)

THE PROBLEM OF CHINA

By Bertrand Russell

CONTENTS

QUESTIONS

CHINA BEFORE THE NINETEENTH CENTURY

CHINA AND THE WESTERN POWERS

MODERN CHINA

JAPAN BEFORE THE RESTORATION

MODERN JAPAN

JAPAN AND CHINA BEFORE 1914

JAPAN AND CHINA DURING THE WAR

THE WASHINGTON CONFERENCE

PRESENT FORCES AND TENDENCIES IN THE FAR EAST

CHINESE AND WESTERN CIVILIZATION CONTRASTED

THE CHINESE CHARACTER

HIGHER EDUCATION IN CHINA

INDUSTRIALISM IN CHINA

THE OUTLOOK FOR CHINA

APPENDIX

INDEX

THE PRACTICE AND THEORY OF BOLSHEVISM

Bertrand Russell

CONTENTS

PAGE

[PART I](#)

THE PRESENT CONDITION OF RUSSIA

I.	WHAT IS HOPED FROM BOLSHEVISM	15
II.	GENERAL CHARACTERISTICS	24
III.	LENIN, TROTSKY AND GORKY	36
IV.	ART AND EDUCATION	45
V.	COMMUNISM AND THE SOVIET CONSTITUTION	72
VI.	THE FAILURE OF RUSSIAN INDUSTRY	81
VII.	DAILY LIFE IN MOSCOW	92
VIII.	TOWN AND COUNTRY	99
IX.	INTERNATIONAL POLICY	106

[PART II](#)

BOLSHEVIK THEORY

I.	THE MATERIALISTIC THEORY OF HISTORY	119
II.	DECIDING FORCES IN POLITICS	128
III.	BOLSHEVIK CRITICISM OF DEMOCRACY	134
IV.	REVOLUTION AND DICTATORSHIP	146
V.	MECHANISM AND THE INDIVIDUAL	157
VI.	WHY RUSSIAN COMMUNISM HAS FAILED	165
VII.	CONDITIONS FOR THE SUCCESS OF COMMUNISM	178

MYSTICISM AND LOGIC

AND OTHER ESSAYS

Bertrand Russell

CONTENTS

<i>Chapter</i>	<i>Page</i>
I. <u>Mysticism and Logic</u>	1
II. <u>The Place of Science in a Liberal Education</u>	33
III. <u>A Free Man's Worship</u>	46
IV. <u>The Study of Mathematics</u>	58
V. <u>Mathematics and the Metaphysicians</u>	74
VI. <u>On Scientific Method in Philosophy</u>	97
VII. <u>The Ultimate Constituents of Matter</u>	125
VIII. <u>The Relation of Sense-data to Physics</u>	145
IX. <u>On the Notion of Cause</u>	180
X. <u>Knowledge by Acquaintance and Knowledge by Description</u>	209
<u>Index</u>	233

OUR KNOWLEDGE OF THE EXTERNAL WORLD AS A FIELD FOR SCIENTIFIC METHOD IN PHILOSOPHY,

By Bertrand Russell

CONTENTS

LECTURE PAGE

I.	Current Tendencies	<u>3</u>
II.	Logic as the Essence of Philosophy	<u>33</u>
III.	On our Knowledge of the External World	<u>63</u>
IV.	The World of Physics and the World of Sense	<u>101</u>
V.	The Theory of Continuity	<u>129</u>
VI.	The Problem of Infinity considered Historically	<u>155</u>
VII.	The Positive Theory of Infinity	<u>185</u>
VIII.	On the Notion of Cause, with Applications to the Free-will Problem	<u>211</u>
	Index	<u>243</u>

INDEX

Absolute, [6](#), [39](#).

Abstraction, principle of, [42](#), [124](#) ff.

Achilles, Zeno's argument of, [173](#).

Acquaintance, [25](#), [144](#).

Activity, [224](#) ff.

Allman, [161 n.](#)

Analysis, [185](#), [204](#), [211](#), [241](#).

legitimacy of, [150](#).

Anaximander, [3](#).

Antinomies, Kant's, [155](#) ff.

Aquinas, [10](#).

Aristotle, [40](#), [160 n.](#), [161](#) ff., [240](#).

Arrow, Zeno's argument of, [173](#).

Assertion, [52](#).

Atomism, logical, [4](#).

Atomists, [160](#).

Belief, [58](#).

primitive and derivative, [69](#) ff.

Bergson, [4](#), [11](#), [13](#), [20](#) ff., [137](#), [138](#), [150](#), [158](#), [165](#), [174](#), [178](#), [229](#) ff.

Berkeley, [63](#), [64](#), [102](#).

Bolzano, [165](#).

Boole, [40](#).

Bradley, [6](#), [39](#), [165](#).

Broad, [172 n.](#)

Brochard, [169 n.](#)

Burnet, [19 n.](#), [160 n.](#), [161 n.](#), [170 n.](#), [171](#) ff.

Calderon, [95](#).

Cantor, [vi](#), [vii](#), [155](#), [165](#), [190](#), [194](#), [199](#).

Categories, [38](#).
Causal laws, [109](#), [212](#) ff.
evidence for, [216](#) ff.
in psychology, [219](#).
Causation, [34](#) ff., [79](#), [212](#) ff.
law of, [221](#).
not *a priori*, [223](#), [232](#).
Cause, [220](#), [223](#).
Certainty, degrees of, [67](#), [68](#), [212](#).
Change,
demands analysis, [151](#).
Cinematograph, [148](#), [174](#).
Classes, [202](#).
non-existence of, [205](#) ff.
Classical tradition, [3](#) ff., [58](#).
Complexity, [145](#), [157](#) ff.
Compulsion, [229](#), [233](#) ff.
Congruence, [195](#).
Consecutiveness, [134](#).
Conservation, [105](#).
Constituents of facts, [51](#), [145](#).
Construction v. inference, iv.
Contemporaries, initial, [119](#), [120 n](#).
Continuity, [64](#), [129](#) ff., [141](#) ff., [155](#) ff.
of change, [106](#), [108](#), [130](#) ff.
Correlation of mental and physical, [233](#).
Counting, [164](#), [181](#), [187](#) ff., [203](#).
Couturat, [40 n](#).
Dante, [10](#).
Darwin, [4](#), [11](#), [23](#), [30](#).
Data, [65](#) ff., [211](#).
“hard” and “soft,” [70](#) ff.
Dates, [117](#).

Definition, [204](#).
Descartes, [5](#), [73](#), [238](#).
Descriptions, [201](#), [214](#).
Desire, [227](#), [235](#).
Determinism, [233](#).
Doubt, [237](#).
Dreams, [85](#), [93](#).
Duration, [146](#), [149](#).
Earlier and later, [116](#).
Effect, [220](#).
Eleatics, [19](#).
Empiricism, [37](#), [222](#).
Enclosure, [114](#) ff., [120](#).
Enumeration, [202](#).
Euclid, [160](#), [164](#).
Evellin, [169](#).
Evolutionism, [4](#), [11](#) ff.
Extension, [146](#), [149](#).
External world, knowledge of, [63](#) ff.
Fact, [51](#).
atomic, [52](#).
Finalism, [13](#).
Form, logical, [42](#) ff., [185](#), [208](#).
Fractions, [132](#), [179](#).
Free will, [213](#), [227](#) ff.
Frege, [5](#), [40](#), [199](#) ff.
Galileo, [4](#), [59](#), [192](#), [194](#), [239](#), [240](#).
Gaye, [169 n.](#), [175](#), [177](#).
Geometry, [5](#).
Giles, [206 n.](#)
Greater and less, [195](#).
Harvard, [4](#).

Hegel, [3](#), [37](#) ff., [46](#), [166](#).
“Here,” [73](#), [92](#).
Hereditary properties, [195](#).
Hippasos, [163](#), [237](#).
Hui Tzu, [206](#).
Hume, [217](#), [221](#).
Hypotheses in philosophy, [239](#).
Illusions, [85](#).
Incommensurables, [162](#) ff., [237](#).
Independence, [73](#), [74](#).
causal and logical, [74](#), [75](#).
Indiscernibility, [141](#), [148](#).
Indivisibles, [160](#).
Induction, [34](#), [222](#).
mathematical, [195](#) ff.
Inductiveness, [190](#), [195](#) ff.
Inference, [44](#), [54](#).
Infinite, [vi](#), [64](#), [133](#), [149](#).
historically considered, [155](#) ff.
“true,” [179](#), [180](#).
positive theory of, [185](#) ff.
Infinitesimals, [135](#).
Instants, [116](#) ff., [129](#), [151](#), [216](#).
defined, [118](#).
Instinct v. Reason, [20](#) ff.
Intellect, [22](#) ff.
Intelligence, how displayed by friends, [93](#).
inadequacy of display, [96](#).
Interpretation, [144](#).
James, [4](#), [10](#), [13](#).
Jourdain, [165 n.](#)
Jowett, [167](#).
Judgment, [58](#).

Kant, [3](#), [112](#), [116](#), [155](#) ff., [200](#).
Knowledge about, [144](#).
Language, bad, [82](#), [135](#).
Laplace, [12](#).
Laws of nature, [218](#) ff.
Leibniz, [13](#), [40](#), [87](#), [186](#), [191](#).
Logic, [201](#).
analytic not constructive, [8](#).
Aristotelian, [5](#).
and fact, [53](#).
inductive, [34](#), [222](#).
mathematical, [vi](#), [40](#) ff.
mystical, [46](#).
and philosophy, [8](#), [33](#) ff., [239](#).
Logical constants, [208](#), [213](#).
Mach, [123](#), [224](#).
Macran, [39 n.](#)
Mathematics, [40](#), [57](#).
Matter, [75](#), [101](#) ff.
permanence of, [102](#) ff.
Measurement, [164](#).
Memory, [230](#), [234](#), [236](#).
Method, deductive, [5](#).
logical-analytic, [v](#), [65](#), [211](#), [236](#) ff.
Milhaud, [168 n.](#), [169 n.](#)
Mill, [34](#), [200](#).
Montaigne, [28](#).
Motion, [130](#), [216](#).
continuous, [133](#), [136](#).
mathematical theory of, [133](#).
perception of, [137](#) ff.
Zeno's arguments on, [168](#) ff.
Mysticism, [19](#), [46](#), [63](#), [95](#).
Newton, [30](#), [146](#).

Nietzsche, [10](#), [11](#).
Noël, [169](#).
Number, cardinal, [131](#), [186](#) ff.
defined, [199](#) ff.
finite, [160](#), [190](#) ff.
inductive, [197](#).
infinite, [178](#), [180](#), [188](#) ff., [197](#).
reflexive, [190](#) ff.
Occam, [107](#), [146](#).
One and many, [167](#), [170](#).
Order, [131](#).
Parmenides, [63](#), [165](#) ff., [178](#).
Past and future, [224](#), [234](#) ff.
Peano, [40](#).
Perspectives, [88](#) ff., [111](#).
Philoponus, [171](#) n.
Philosophy and ethics, [26](#) ff.
and mathematics, [185](#) ff.
province of, [17](#), [26](#), [185](#), [236](#).
scientific, [11](#), [16](#), [18](#), [29](#), [236](#) ff.
Physics, [101](#) ff., [147](#), [239](#), [242](#).
descriptive, [224](#).
verifiability of, [81](#), [110](#).
Place, [86](#), [90](#).
at and *from*, [92](#).
Plato, [4](#), [19](#), [27](#), [46](#), [63](#), [165](#) n., [166](#), [167](#).
Poincaré, [123](#), [141](#).
Points, [113](#) ff., [129](#), [158](#).
definition of, [vi](#), [115](#).
Pragmatism, [11](#).
Prantl, [174](#).
Predictability, [229](#) ff.
Premisses, [211](#).

Probability, [36](#).
Propositions, [52](#).
 atomic, [52](#).
 general, [55](#).
 molecular, [54](#).
Pythagoras, [19](#), [160](#) ff., [237](#).
Race-course, Zeno's argument of, [171](#) ff.
Realism, new, [6](#).
Reflexiveness, [190](#) ff.
Relations, [45](#).
 asymmetrical, [47](#).
 Bradley's reasons against, [6](#).
 external, [150](#).
 intransitive, [48](#).
 multiple, [50](#).
 one-one, [203](#).
 reality of, [49](#).
 symmetrical, [47](#), [124](#).
 transitive, [48](#), [124](#).
Relativity, [103](#), [242](#).
Repetitions, [230](#) ff.
Rest, [136](#).
Ritter and Preller, [161](#) n.
Robertson, D. S., [160](#) n.
Rousseau, [20](#).
Royce, [50](#).
Santayana, [46](#).
Scepticism, [66](#), [67](#).
Seeing double, [86](#).
Self, [73](#).
Sensation, [25](#), [75](#), [123](#).
 and stimulus, [139](#).
Sense-data, [56](#), [63](#), [67](#), [75](#), [110](#), [141](#), [143](#), [213](#).
 and physics, [v](#), [64](#), [81](#), [97](#), [101](#) ff., [140](#).

infinitely numerous? [149](#), [159](#).
Sense-perception, [53](#).
Series, [49](#).
compact, [132](#), [142](#), [178](#).
continuous, [131](#), [132](#).
Sigwart, [187](#).
Simplicius, [170 n](#).
Simultaneity, [116](#).
Space, [73](#), [88](#), [103](#), [112](#) ff., [130](#).
absolute and relative, [146](#), [159](#).
antinomies of, [155](#) ff.
perception of, [68](#).
of perspectives, [88](#) ff.
private, [89](#), [90](#).
of touch and sight, [78](#), [113](#).
Spencer, [4](#), [12](#), [236](#).
Spinoza, [46](#), [166](#).
Stadium, Zeno's argument of, [134 n](#), [175](#) ff.
Subject-predicate, [45](#).
Synthesis, [157](#), [185](#).
Tannery, Paul, [169 n](#).
Teleology, [223](#).
Testimony, [67](#), [72](#), [82](#), [87](#), [96](#), [212](#).
Thales, [3](#).
Thing-in-itself, [75](#), [84](#).
Things, [89](#) ff., [104](#) ff., [213](#).
Time, [103](#), [116](#) ff., [130](#), [155](#) ff., [166](#), [215](#).
absolute or relative, [146](#).
local, [103](#).
private, [121](#).
Uniformities, [217](#).
Unity, organic, [9](#).
Universal and particular, [39 n](#).

Volition, [223](#) ff.

Whitehead, [vi](#), [207](#).

Wittgenstein, [vii](#), [208 n](#).

Worlds, actual and ideal, [111](#).

possible, [186](#).

private, [88](#).

Zeller, [173](#).

Zeno, [129](#), [134](#), [136](#), [165](#) ff.

[1] Delivered as Lowell Lectures in Boston, in March and April 1914.

[2] London and New York, 1912 (“Home University Library”).

[3] The first volume was published at Cambridge in 1910, the second in 1912, and the third in 1913.

[4] *Appearance and Reality*, pp. 32–33.

[5] *Creative Evolution*, English translation, p. 41.

[6] Cf. Burnet, *Early Greek Philosophy*, pp. 85 ff.

[7] *Introduction to Metaphysics*, p. 1.

[8] *Logic*, book iii., chapter iii., § 2.

[9] Book iii., chapter xxi., § 3.

[10] Or rather a propositional function.

[11] The subject of causality and induction will be discussed again in [Lecture VIII](#).

[12] See the translation by H. S. Macran, *Hegel's Doctrine of Formal Logic*, Oxford, 1912. Hegel's argument in this portion of his “Logic” depends throughout upon confusing the “is” of predication, as in “Socrates is mortal,” with the “is” of identity, as in “Socrates is the philosopher who drank the hemlock.” Owing to this confusion, he thinks that “Socrates” and “mortal” must be identical. Seeing that they are different, he does not infer, as others would, that there is a mistake somewhere, but that they exhibit “identity in difference.” Again, Socrates is particular, “mortal” is universal. Therefore, he says, since Socrates is mortal, it follows that the particular is the universal—taking the “is” to be throughout expressive of identity. But to say “the particular is the universal” is self-contradictory. Again Hegel does not suspect a mistake but proceeds to synthesise particular and universal in the individual, or

concrete universal. This is an example of how, for want of care at the start, vast and imposing systems of philosophy are built upon stupid and trivial confusions, which, but for the almost incredible fact that they are unintentional, one would be tempted to characterise as puns.

[13] Cf. Couturat, *La Logique de Leibniz*, pp. 361, 386.

[14] It was often recognised that there was *some* difference between them, but it was not recognised that the difference is fundamental, and of very great importance.

[15] *Encyclopædia of the Philosophical Sciences*, vol. i. p. 97.

[16] This perhaps requires modification in order to include such facts as beliefs and wishes, since such facts apparently contain propositions as components. Such facts, though not strictly atomic, must be supposed included if the statement in the text is to be true.

[17] The assumptions made concerning time-relations in the above are as follows:—

[18] The above paradox is essentially the same as Zeno's argument of the stadium which will be considered in our [next lecture](#).

[19] See [next lecture](#).

[20] *Monist*, July 1912, pp. 337–341.

[21] “Le continu mathématique,” *Revue de Métaphysique et de Morale*, vol. i. p. 29.

[22] In what concerns the early Greek philosophers, my knowledge is largely derived from Burnet's valuable work, *Early Greek Philosophy* (2nd ed., London, 1908). I have also been greatly assisted by Mr D. S. Robertson of Trinity College, who has supplied the deficiencies of my knowledge of Greek, and brought important references to my notice.

[23] Cf. Aristotle, *Metaphysics*, M. 6, 1080b, 18 *sqq.*, and 1083b, 8 *sqq.*

[24] There is some reason to think that the Pythagoreans distinguished between discrete and continuous quantity. G. J. Allman, in his *Greek Geometry from Thales to Euclid*, says (p. 23): “The Pythagoreans made a fourfold division of mathematical science, attributing one of its parts to the how many, τὸ πῶς, and the other to the how much, τὸ πόθεν; and they assigned to each of these parts a twofold division. For they said that discrete quantity, or the *how many*, either subsists by itself or must be considered with relation to some other; but that continued quantity, or the

how much, is either stable or in motion. Hence they affirmed that arithmetic contemplates that discrete quantity which subsists by itself, but music that which is related to another; and that geometry considers continued quantity so far as it is immovable; but astronomy (t? sfa?????) contemplates continued quantity so far as it is of a self-motive nature. (Proclus, ed. Friedlein, p. 35. As to the distinction between t? p?????, continuous, and t? p?s??, discrete quantity, see Iambl., in *Nicomachi Geraseni Arithmetica introductionem*, ed. Tennulius, p. 148.)” Cf. p. 48.

[25] Referred to by Burnet, *op. cit.*, p. 120.

[26] iv., 6. 213b, 22; H. Ritter and L. Preller, *Historia Philosophiæ Græcæ*, 8th ed., Gotha, 1898, p. 75 (this work will be referred to in future as “R. P.”).

[27] The Pythagorean proof is roughly as follows. If possible, let the ratio of the diagonal to the side of a square be m/n , where m and n are whole numbers having no common factor. Then we must have $m^2 = 2n^2$. Now the square of an odd number is odd, but m^2 , being equal to $2n^2$, is even. Hence m must be even. But the square of an even number divides by 4, therefore n^2 , which is half of m^2 , must be even. Therefore n must be even. But, since m is even, and m and n have no common factor, n must be odd. Thus n must be both odd and even, which is impossible; and therefore the diagonal and the side cannot have a rational ratio.

[28] In regard to Zeno and the Pythagoreans, I have derived much valuable information and criticism from Mr P. E. B. Jourdain.

[29] So Plato makes Zeno say in the *Parmenides*, apropos of his philosophy as a whole; and all internal and external evidence supports this view.

[30] “With Parmenides,” Hegel says, “philosophising proper began.” *Werke* (edition of 1840), vol. xiii. p. 274.

[31] *Parmenides*, 128 A–D.

[32] This interpretation is combated by Milhaud, *Les philosophes-géomètres de la Grèce*, p. 140 n., but his reasons do not seem to me convincing. All the interpretations in what follows are open to question, but all have the support of reputable authorities.

[33] *Physics*, vi. 9. 239b (R.P. 136–139).

[34] Cf. Gaston Milhaud, *Les philosophes-géomètres de la Grèce*, p. 140 n.; Paul Tannery, *Pour l'histoire de la science hellène*, p. 249; Burnet, *op. cit.*, p. 362.

[35] Cf. R. K. Gaye, "On Aristotle, *Physics*, Z ix." *Journal of Philology*, vol. xxxi., esp. p. 111. Also Moritz Cantor, *Vorlesungen über Geschichte der Mathematik*, 1st ed., vol. i., 1880, p. 168, who, however, subsequently adopted Paul Tannery's opinion, *Vorlesungen*, 3rd ed. (vol. i. p. 200).

[36] "Le mouvement et les partisans des indivisibles," *Revue de Métaphysique et de Morale*, vol. i. pp. 382–395.

[37] "Le mouvement et les arguments de Zénon d'Élée," *Revue de Métaphysique et de Morale*, vol. i. pp. 107–125.

[38] Cf. M. Brochard, "Les prétendus sophismes de Zénon d'Élée," *Revue de Métaphysique et de Morale*, vol. i. pp. 209–215.

[39] Simplicius, *Phys.*, 140, 28 D (R.P. 133); Burnet, *op. cit.*, pp. 364–365.

[40] *Op. cit.*, p. 367.

[41] Aristotle's words are: "The first is the one on the non-existence of motion on the ground that what is moved must always attain the middle point sooner than the end-point, on which we gave our opinion in the earlier part of our discourse." *Phys.*, vi. 9. 939B (R.P. 136). Aristotle seems to refer to *Phys.*, vi. 2. 223AB [R.P. 136A]: "All space is continuous, for time and space are divided into the same and equal divisions.... Wherefore also Zeno's argument is fallacious, that it is impossible to go through an infinite collection or to touch an infinite collection one by one in a finite time. For there are two senses in which the term 'infinite' is applied both to length and to time, and in fact to all continuous things, either in regard to divisibility, or in regard to the ends. Now it is not possible to touch things infinite in regard to number in a finite time, but it is possible to touch things infinite in regard to divisibility: for time itself also is infinite in this sense. So that in fact we go through an infinite, [space] in an infinite [time] and not in a finite [time], and we touch infinite things with infinite things, not with finite things." Philoponus, a sixth-century commentator (R.P. 136A, *Exc. Paris Philop. in Arist. Phys.*, 803, 2. Vit.), gives the following illustration: "For if a thing were moved the space of a cubit in one hour, since in every space there are an infinite number of points, the thing moved must needs

touch all the points of the space: it will then go through an infinite collection in a finite time, which is impossible.”

[42] Cf. Mr C. D. Broad, “Note on Achilles and the Tortoise,” *Mind*, N.S., vol. xxii. pp. 318–9.

[43] *Op. cit.*

[44] Aristotle's words are: “The second is the so-called Achilles. It consists in this, that the slower will never be overtaken in its course by the quickest, for the pursuer must always come first to the point from which the pursued has just departed, so that the slower must necessarily be always still more or less in advance.” *Phys.*, vi. 9. 239B (R.P. 137).

[45] *Phys.*, vi. 9. 239B (R.P. 138).

[46] *Phys.*, vi. 9. 239B (R.P. 139).

[47] *Loc. cit.*

[48] *Loc. cit.*, p. 105.

[49] *Phil. Werke*, Gerhardt's edition, vol. i. p. 338.

[50] *Mathematical Discourses concerning two new sciences relating to mechanics and local motion, in four dialogues*. By Galileo Galilei, Chief Philosopher and Mathematician to the Grand Duke of Tuscany. Done into English from the Italian, by Tho. Weston, late Master, and now published by John Weston, present Master, of the Academy at Greenwich. See pp. 46 ff.

[51] In his *Grundlagen einer allgemeinen Mannichfaltigkeitslehre* and in articles in *Acta Mathematica*, vol. ii.

[52] The definition of number contained in this book, and elaborated in the *Grundgesetze der Arithmetik* (vol. i., 1893; vol. ii., 1903), was rediscovered by me in ignorance of Frege's work. I wish to state as emphatically as possible—what seems still often ignored—that his discovery antedated mine by eighteen years.

[53] Giles, *The Civilisation of China* (Home University Library), p. 147.

[54] Cf. *Principia Mathematica*, § 20, and Introduction, chapter iii.

[55] In the above remarks I am making use of unpublished work by my friend Ludwig Wittgenstein.

[56] Thus we are not using “thing” here in the sense of a class of correlated “aspects,” as we did in [Lecture III](#). Each “aspect” will count

separately in stating causal laws.

[\[57\]](#) The above remarks, for purposes of illustration, adopt one of several possible opinions on each of several disputed points.

AN ESSAY ON THE FOUNDATIONS OF GEOMETRY

By Bertrand A. W. Russell

Fellow Of Trinity College, Cambridge

1897

CONTENTS

INTRODUCTION.

OUR PROBLEM DEFINED BY ITS RELATIONS TO
LOGIC, PSYCHOLOGY AND MATHEMATICS.

	PAGE
<u>1.</u> The problem first received a modern form through Kant, who connected the <i>à priori</i> with the subjective	1
<u>2.</u> A mental state is subjective, for Psychology, when its immediate cause does not lie in the outer world	2
<u>3.</u> A piece of knowledge is <i>à priori</i> , for Epistemology, when without it knowledge would be impossible	2
<u>4.</u> The subjective and the <i>à priori</i> belong respectively to Psychology and to Epistemology. The latter alone will be investigated in this essay	3
<u>5.</u> My test of the <i>à priori</i> will be purely logical: what knowledge is necessary for experience?	3
<u>6.</u> But since the necessary is hypothetical, we must include, in the <i>à priori</i> , the ground of necessity	4
<u>7.</u> This may be the essential postulate of our science, or the element, in the subject-matter, which is necessary to experience;	4
<u>8.</u> Which, however, are both at bottom the same ground	5
<u>9.</u> Forecast of the work	5

CHAPTER I.

A SHORT HISTORY OF METAGEOMETRY.

- [10.](#) Metageometry began by rejecting the axiom of parallels 7
- [11.](#) Its history may be divided into three periods: the synthetic, the 7
metrical and the projective
- [12.](#) The first period was inaugurated by Gauss, 10
- [13.](#) Whose suggestions were developed independently by 10
Lobatchewsky
- [14.](#) And Bolyai 11
- The purpose of all three was to show that the axiom of
- [15.](#) parallels could not be deduced from the others, since its denial 12
did not lead to contradictions
- [16.](#) The second period had a more philosophical aim, and was 13
inspired chiefly by Gauss and Herbart
- [17.](#) The first work of this period, that of Riemann, invented two 14
new conceptions:
- [18.](#) The first, that of a manifold, is a class-conception, containing 14
space as a species,
- [19.](#) And defined as such that its determinations form a collection 15
of magnitudes
- [20.](#) The second, the measure of curvature of a manifold, grew out 16
of curvature in curves and surfaces
- [21.](#) By means of Gauss's analytical formula for the curvature of 19
surfaces,
- Which enables us to define a *constant* measure of curvature of
- [22.](#) a three-dimensional space without reference to a fourth 20
dimension
- The main result of Riemann's mathematical work was to show
- [23.](#) that, if magnitudes are independent of place, the measure of 21
curvature of space must be constant
- [24.](#) Helmholtz, who was more of a philosopher than a 22
mathematician,
- [25.](#) Gave a new but incorrect formulation of the essential axioms, 23

- [26.](#) And deduced the quadratic formula for the infinitesimal arc, which Riemann had assumed 24
- [27.](#) Beltrami gave Lobatchewsky's planimetry a Euclidean interpretation, 25
- [28.](#) Which is analogous to Cayley's theory of distance; 26
- [29.](#) And dealt with n -dimensional spaces of constant negative curvature 27
- [30.](#) The third period abandons the metrical methods of the second, and extrudes the notion of spatial quantity 27
- [31.](#) Cayley reduced metrical properties to projective properties, relative to a certain conic or quadric, the Absolute; 28
- [32.](#) And Klein showed that the Euclidean or non-Euclidean systems result, according to the nature of the Absolute; 29
- [33.](#) Hence Euclidean *space* appeared to give rise to all the kinds of Geometry, and the question, which is true, appeared reduced to one of convention 30
- [34.](#) But this view is due to a confusion as to the nature of the coordinates employed 30
- [35.](#) Projective coordinates have been regarded as dependent on distance, and thus really metrical 31
- [36.](#) But this is not the case, since anharmonic ratio can be projectively defined 32
- [37.](#) Projective coordinates, being purely descriptive, can give no information as to metrical properties, and the reduction of metrical to projective properties is purely technical 33
- [38.](#) The true connection of Cayley's measure of distance with non-Euclidean Geometry is that suggested by Beltrami's Saggio, and worked out by Sir R. Ball, 36
- [39.](#) Which provides a Euclidean equivalent for every non-Euclidean proposition, and so removes the possibility of contradictions in Metageometry 38
- [40.](#) Klein's elliptic Geometry has not been proved to have a corresponding variety of space 39

- [41.](#) The geometrical use of imaginaries, of which Cayley 41
demanded a philosophical discussion,
- [42.](#) Has a merely technical validity, 42
- [43.](#) And is capable of giving geometrical results only when it 45
begins and ends with real points and figures
- [44.](#) We have now seen that projective Geometry is logically prior 46
to metrical Geometry, but cannot supersede it
- [45.](#) Sophus Lie has applied projective methods to Helmholtz's 46
formulation of the axioms, and has shown the axiom of
Monodromy to be superfluous
- [46.](#) Metageometry has gradually grown independent of 50
philosophy, but has grown continually more interesting to
philosophy
- [47.](#) Metrical Geometry has three indispensable axioms, 50
- [48.](#) Which we shall find to be not results, but conditions, of 51
measurement,
- [49.](#) And which are nearly equivalent to the three axioms of 52
projective Geometry
- [50.](#) Both sets of axioms are necessitated, not by facts, but by logic 52

CHAPTER II.

CRITICAL ACCOUNT OF SOME PREVIOUS PHILOSOPHICAL THEORIES OF GEOMETRY.

- [51.](#) A criticism of representative modern theories need not begin 54
before Kant
- [52.](#) Kant's doctrine must be taken, in an argument about 55
Geometry, on its purely logical side
- [53.](#) Kant contends that since Geometry is apodeictic, space must 55
be *à priori* and subjective, while since space is *à priori* and
subjective, Geometry must be apodeictic
- [54.](#) Metageometry has upset the first line of argument, not the 56
second
- [55.](#) The second may be attacked by criticizing either the 57
distinction of synthetic and analytic judgments, or the first two

	arguments of the metaphysical deduction of space	
56.	Modern Logic regards every judgment as both synthetic and analytic,	57
57.	But leaves the <i>à priori</i> , as that which is presupposed in the possibility of experience	59
58.	Kant's first two arguments as to space suffice to prove <i>some</i> form of externality, but not necessarily Euclidean space, a necessary condition of experience	60
59.	Among the successors of Kant, Herbart alone advanced the theory of Geometry, by influencing Riemann	62
60.	Riemann regarded space as a particular kind of manifold, i.e. wholly quantitatively	63
61.	He therefore unduly neglected the qualitative adjectives of space	64
62.	His philosophy rests on a vicious disjunction	65
63.	His definition of a manifold is obscure,	66
64.	And his definition of measurement applies only to space	67
65.	Though mathematically invaluable, his view of space as a manifold is philosophically misleading	69
66.	Helmholtz attacked Kant both on the mathematical and on the psychological side;	70
67.	But his criterion of apriority is changeable and often invalid;	71
68.	His proof that non-Euclidean spaces are imaginable is inconclusive;	72
69.	And his assertion of the dependence of measurement on rigid bodies, which may be taken in three senses,	74
70.	Is wholly false if it means that the axiom of Congruence actually asserts the existence of rigid bodies,	75
71.	Is untrue if it means that the necessary reference of geometrical propositions to matter renders pure Geometry empirical,	76
72.	And is inadequate to his conclusion if it means, what is true, that <i>actual</i> measurement involves approximately rigid bodies	78

- Geometry deals with an abstract matter, whose physical
[73.](#) properties are disregarded; and Physics must presuppose 80
Geometry
- [74.](#) Erdmann accepted the conclusions of Riemann and 81
Helmholtz,
- [75.](#) And regarded the axioms as necessarily successive steps in 82
classifying space as a species of manifold
- [76.](#) His deduction involves four fallacious assumptions, namely: 82
- [77.](#) That conceptions must be abstracted from a series of 83
instances;
- [78.](#) That all definition is classification; 83
- [79.](#) That conceptions of magnitude can be applied to space as a 84
whole;
- [80.](#) And that if conceptions of magnitude could be so applied, all 86
the adjectives of space would result from their application
- [81.](#) Erdmann regards Geometry alone as incapable of deciding on 86
the truth of the axiom of Congruence,
- [82.](#) Which he affirms to be empirically proved by Mechanics. 88
- [83.](#) The variety and inadequacy of Erdmann's tests of apriority 89
- [84.](#) Invalidate his final conclusions on the theory of Geometry 90
- [85.](#) Lotze has discussed two questions in the theory of Geometry: 93
- [86.](#) (1) He regards the possibility of non-Euclidean spaces as 93
suggested by the subjectivity of space,
- [87.](#) And rejects it owing to a mathematical misunderstanding, 96
- [88.](#) Having missed the most important sense of their possibility, 96
- [89.](#) Which is that they fulfil the logical conditions to which any 97
form of externality must conform
- [90.](#) (2) He attacks the mathematical procedure of Metageometry 98
- [91.](#) The attack begins with a question-begging definition of 99
parallels
- [92.](#) Lotze maintains that all apparent departures from Euclid could
be physically explained, a view which really makes Euclid 99
empirical

<u>93.</u>	His criticism of Helmholtz's analogies rests wholly on mathematical mistakes	101
<u>94.</u>	His proof that space must have three dimensions rests on neglect of different orders of infinity	104
<u>95.</u>	He attacks non-Euclidean spaces on the mistaken ground that they are not homogeneous	107
<u>96.</u>	Lotze's objections fall under four heads	108
<u>97.</u>	Two other semi-philosophical objections may be urged,	109
<u>98.</u>	One of which, the absence of similarity, has been made the basis of attack by Delbouf,	110
<u>99.</u>	But does not form a valid ground of objection	111
<u>100.</u>	Recent French speculation on the foundations of Geometry has suggested few new views	112
<u>101.</u>	All homogeneous spaces are <i>à priori</i> possible, and the decision between them is empirical	114

CHAPTER III.

Section A. the axioms of projective geometry.

<u>102.</u>	Projective Geometry does not deal with magnitude, and applies to all spaces alike	117
<u>103.</u>	It will be found wholly <i>à priori</i>	117
<u>104.</u>	Its axioms have not yet been formulated philosophically	118
<u>105.</u>	Coordinates, in projective Geometry, are not spatial magnitudes, but convenient names for points	118
<u>106.</u>	The possibility of distinguishing various points is an axiom	119
<u>107.</u>	The qualitative relations between points, dealt with by projective Geometry, are presupposed by the quantitative treatment	119
<u>108.</u>	The only qualitative relation between two points is the straight line, and all straight lines are qualitatively similar	120
<u>109.</u>	Hence follows, by extension, the principle of projective transformation	121

<u>110.</u>	By which figures qualitatively indistinguishable from a given figure are obtained	122
<u>111.</u>	Anharmonic ratio may and must be descriptively defined	122
<u>112.</u>	The quadrilateral construction is essential to the projective definition of points,	123
<u>113.</u>	And can be projectively defined,	124
<u>114.</u>	By the general principle of projective transformation	126
<u>115.</u>	The principle of duality is the mathematical form of a philosophical circle,	127
<u>116.</u>	Which is an inevitable consequence of the relativity of space, and makes any definition of the point contradictory	128
<u>117.</u>	We define the point as that which is spatial, but contains no space, whence other definitions follow	128
<u>118.</u>	What is meant by qualitative equivalence in Geometry?	129
<u>119.</u>	Two pairs of points on one straight line, or two pairs of straight lines through one point, are qualitatively equivalent	129
<u>120.</u>	This explains why <i>four</i> collinear points are needed, to give an intrinsic relation by which the fourth can be descriptively defined when the first three are given	130
<u>121.</u>	Any two projectively related figures are qualitatively equivalent, i.e. differ in no non-quantitative conceptual property	131
<u>122.</u>	Three axioms are used by projective Geometry,	132
<u>123.</u>	And are required for qualitative spatial comparison,	132
<u>124.</u>	Which involves the homogeneity, relativity and passivity of space	133
<u>125.</u>	The conception of a form of externality,	134
<u>126.</u>	Being a creature of the intellect, can be dealt with by pure mathematics	134
<u>127.</u>	The resulting doctrine of extension will be, for the moment, hypothetical	135
<u>128.</u>	But is rendered assertorical by the necessity, for experience, of some form of externality	136
<u>129.</u>	Any such form must be relational	136

130.	And homogeneous	137
131.	And the relations constituting it must appear infinitely divisible	137
132.	It must have a finite integral number of dimensions,	139
133.	Owing to its passivity and homogeneity	140
134.	And to the systematic unity of the world	140
135.	A one-dimensional form alone would not suffice for experience	141
136.	Since its elements would be immovably fixed in a series	142
137.	Two positions have a relation independent of other positions,	143
138.	Since positions are wholly defined by mutually independent relations	143
139.	Hence projective Geometry is wholly <i>à priori</i> ,	146
140.	Though metrical Geometry contains an empirical element	146
	Section B. the axioms of metrical geometry.	
141.	Metrical Geometry is distinct from projective, but has the same fundamental postulate	147
142.	It introduces the new idea of motion, and has three <i>à priori</i> axioms	148
	I. <i>The Axiom of Free Mobility.</i>	
143.	Measurement requires a criterion of spatial equality	149
144.	Which is given by superposition, and involves the axiom of Free Mobility	150
145.	The denial of this axiom involves an action of empty space on things	151
146.	There is a mathematically possible alternative to the axiom,	152
147.	Which, however, is logically and philosophically untenable	153
148.	Though Free Mobility is <i>à priori</i> , actual measurement is empirical	154
149.	Some objections remain to be answered, concerning—	154
150.	(1) The comparison of volumes and of Kant's symmetrical objects	154
151.	(2) The measurement of time, where congruence is impossible	156

152.	(3) The immediate perception of spatial magnitude; and	157
153.	(4) The Geometry of non-congruent surfaces	158
154.	Free Mobility includes Helmholtz's Monodromy	159
155.	Free Mobility involves the relativity of space	159
156.	From which, reciprocally, it can be deduced	160
157.	Our axiom is therefore <i>à priori</i> in a double sense	160

II. *The Axiom of Dimensions.*

158.	Space must have a finite integral number of dimensions	161
159.	But the restriction to three is empirical	162
160.	The general axiom follows from the relativity of position	162
161.	The limitation to three dimensions, unlike most empirical knowledge, is accurate and certain	163

III. *The Axiom of Distance.*

162.	The axiom of distance corresponds, here, to that of the straight line in projective Geometry	164
163.	The possibility of spatial measurement involves a magnitude uniquely determined by two points,	164
164.	Since two points must have some relation, and the passivity of space proves this to be independent of external reference	165
165.	There can be only one such relation	166
166.	This must be measured by a curve joining the two points,	166
167.	And the curve must be uniquely determined by the two points	167
168.	Spherical Geometry contains an exception to this axiom,	168
169.	Which, however, is not quite equivalent to Euclid's	168
170.	The exception is due to the fact that two points, in spherical space, may have an external relation unaltered by motion,	169
171.	Which, however, being a relation of linear magnitude, presupposes the possibility of linear magnitude	170
172.	A relation between two points must be a line joining them	170
173.	Conversely, the existence of a unique line between two points can be deduced from the nature of a form of externality,	171

- [174.](#) And necessarily leads to distance, when quantity is applied to it 172
- [175.](#) Hence the axiom of distance, also, is *à priori* in a double sense 172
- [176.](#) No metrical coordinate system can be set up without the straight line 174
- [177.](#) No axioms besides the above three are necessary to metrical Geometry 175
- [178.](#) But these three are necessary to the direct measurement of any continuum 176
- [179.](#) Two philosophical questions remain for a final chapter 177

CHAPTER IV.

PHILOSOPHICAL CONSEQUENCES.

- [180.](#) What is the relation to experience of a form of externality in general? 178
- This form is the class-conception, containing every possible
- [181.](#) intuition of externality; and some such intuition is necessary to experience 178
- [182.](#) What relation does this view bear to Kant's? 179
- [183.](#) It is less psychological, since it does not discuss whether space is given in sensation, 180
- And maintains that not only space, but any form of externality
- [184.](#) which renders experience possible, must be given in sense-perception 181
- [185.](#) Externality should mean, not externality to the Self, but the mutual externality of presented things 181
- [186.](#) Would this be unknowable without a given form of externality? 182
- [187.](#) Bradley has proved that space and time preclude the existence of mere particulars, 182
- [188.](#) And that knowledge requires the *This* to be neither simple nor self-subsistent 183
- [189.](#) To prove that experience requires a form of externality, I assume that all knowledge requires the recognition of identity 184

in difference

[190.](#) Such recognition involves time 184

[191.](#) And some other form giving simultaneous diversity 185

The above argument has not deduced sense-perception from
[192.](#) the categories, but has shown the former, unless it contains a 186
certain element, to be unintelligible to the latter

[193.](#) How to account for the realization of this element, is a 187
question for metaphysics

[194.](#) What are we to do with the contradictions in space? 188

[195.](#) Three contradictions will be discussed in what follows 188

[196.](#) (1) The antinomy of the Point proves the relativity of space, 189

[197.](#) And shows that Geometry must have some reference to 190
matter,

[198.](#) By which means it is made to refer to spatial order, not to 191
empty space

The causal properties of matter are irrelevant to Geometry,
[199.](#) which must regard it as composed of unextended atoms, by 191
which points are replaced

[200.](#) (2) The circle in defining straight lines and planes is overcome 192
by the same reference to matter

[201.](#) (3) The antinomy that space is relational and yet more than 193
relational,

[202.](#) Seems to depend on the confusion of empty space with spatial 193
order

[203.](#) Kant regarded empty space as the subject-matter of Geometry, 194

[204.](#) But the arguments of the Aesthetic are inconclusive on this 195
point,

[205.](#) And are upset by the mathematical antinomies, which prove 196
that spatial order should be the subject-matter of Geometry

[206.](#) The apparent thinghood of space is a psychological illusion, 196
due to the fact that spatial relations are immediately given

The apparent divisibility of spatial relations is either an
[207.](#) illusion, arising out of empty space, or the expression of the 197
possibility of quantitatively different spatial relations

[208.](#) Externality is not a relation, but an aspect of relations. Spatial order, owing to its reference to matter, is a real relation 198

[209.](#) Conclusion 199

WHY MEN FIGHT

A METHOD OF ABOLISHING THE INTERNATIONAL DUEL

By Bertrand Russell

CONTENTS

CHAPTER	PAGE
I The Principle of Growth	<u>3</u>
II The State	<u>42</u>
III War as an Institution	<u>79</u>
IV Property	<u>117</u>
V Education	<u>153</u>
VI Marriage and the Population Question	<u>182</u>
VII Religion and the Churches	<u>215</u>
VIII What We Can Do	<u>245</u>

*** END OF THE PROJECT GUTENBERG EBOOK INDEX OF THE
PROJECT GUTENBERG WORKS OF BERTRAND RUSSELL ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

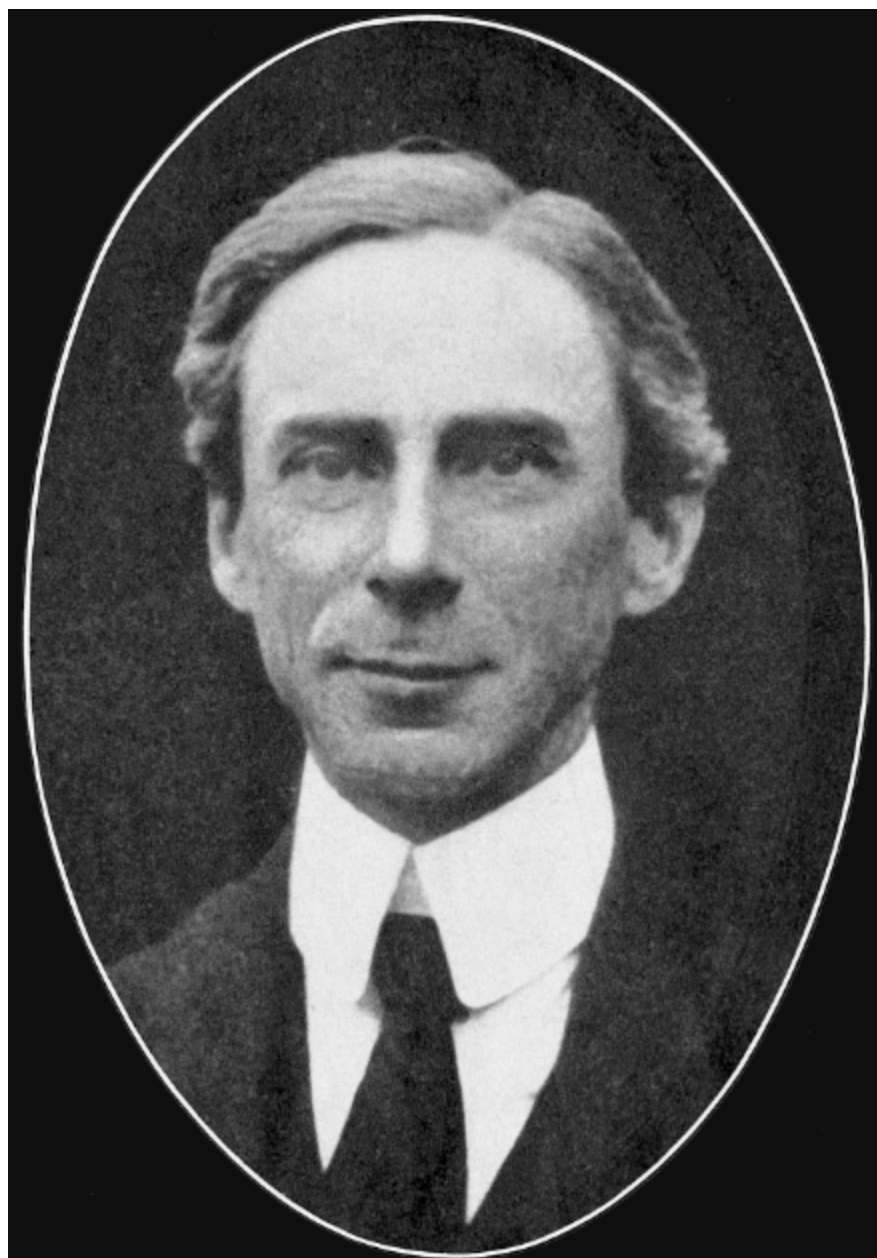
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



[back](#)